EECE 306 Lab 9. Current Sources and the Biot-Savart Law

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This notebook checks current sources against a circular loop, a straight wire, and a solenoid. All positions use meters and H uses A/m. The experiments below run again each time the notebook is published.

Setup and implementation time

The initial AI-assisted implementation and first loop check took about 1.2 minutes of wall-clock time. That number covers the first code pass, not the team's working time or the full lab. The distance calculation, inverse-cube factor, and batches of observation points came from the Lab 3 electric field engine, which needed no changes. H replaces the charge numerator with Idl cross displacement and uses 1/(4*pi) at the end. Lab 4 had kept only the tangent's length, so its curve function now returns the tangent and node weights too; keeping that direction earlier would have saved this extra step.

```
Initial AI-assisted implementation/check: 1.22 minutes
GNU Octave 10.3.0
```

Circular loop and symmetry

A counterclockwise loop of radius 1 m carries 1 A. Its on-axis field is `I*a^2/(2*(a^2+z^2)^(3/2))`, with `I/(2*a)` at the center. The ten relative errors below must stay below 1e-6. The transverse components should be zero, so their absolute tolerance is 1e-12 times the 0.5 A/m center field.

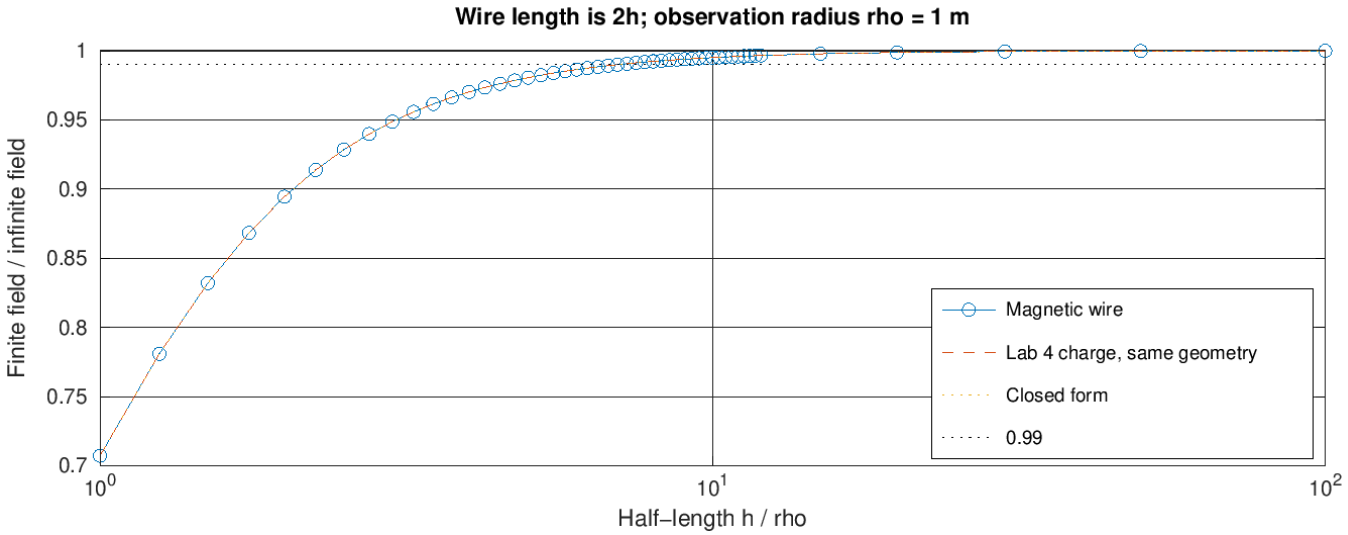
```
 z (m)    H_z (A/m)    relative error
 0.000     0.5000000     1.040e-11
 0.222     0.4651244     1.040e-11
 0.444     0.3815399     1.040e-11
 0.667     0.2880174     1.040e-11
 0.889     0.2087592     1.040e-11
 1.111     0.1496855     1.040e-11
 1.333     0.1080000     1.040e-11
 1.556     0.0790638     1.040e-11
 1.778     0.0589186     1.040e-11
 2.000     0.0447214     1.040e-11
Center 0.500000000 A/m; expected 0.500000000 A/m
Max transverse 1.974e-14 A/m; absolute tolerance 5.0e-13 A/m
```

Finite wire and the Lab 4 comparison

Here h is the wire's half-length, so its endpoints are -h and h. The normalized exact field is (h/rho)/sqrt(1+(h/rho)^2). The finer sweep brackets the 99 percent crossing instead of treating one coarse sample as an exact threshold.

The electric and magnetic thresholds agree for the same total length because their radial integrands have the same distance dependence. The refined wire first passes at h/rho = 7.25, or full length/rho = 14.5. Lab 4 first passed at 19.31 on its much wider sample spacing. Using the same geometry and grid removes that sampling difference; the analytic full-length threshold is about 14.04 in both cases.

```
Wire h/rho 7.00 -> ratio 0.98994951
Wire h/rho 7.25 -> ratio 0.99062114 (first >= 0.99)
Analytic half-length crossing 7.017924; full-length crossing 14.035848
Lab 4 original coarse full-length sample: L/rho 19.3070
Max electric/magnetic ratio difference on the same grid: 3.997e-15
```

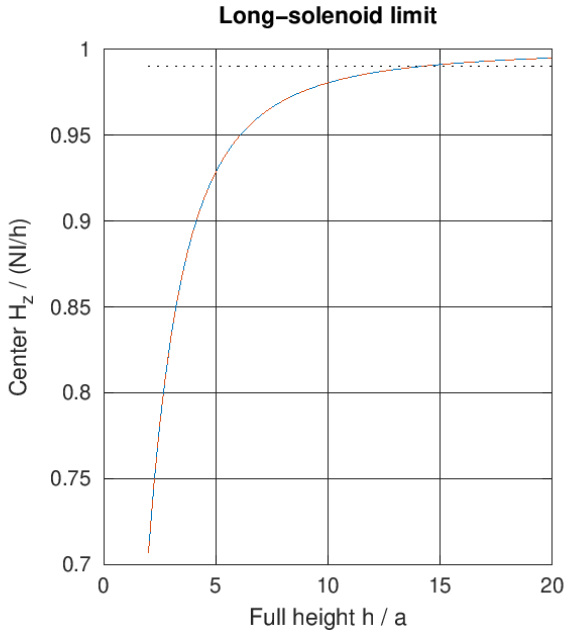


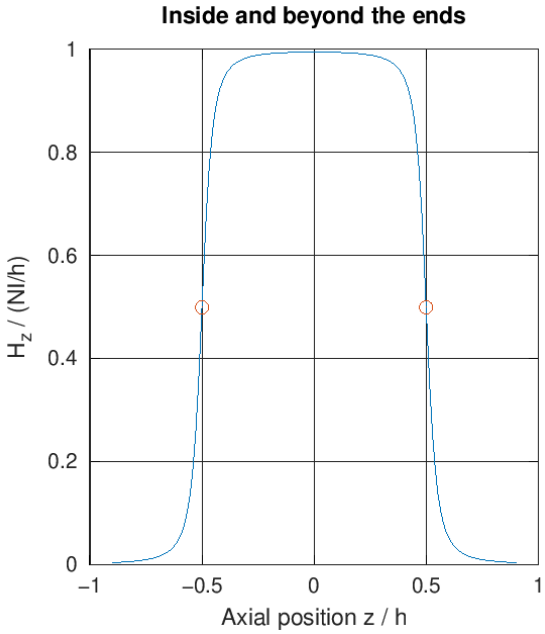
Solenoid aspect ratio and axial profile

This model uses 200 loops with turn centers halfway through equal height intervals. Its density is then N/h, matching the reference formula. A fixed stack of 25 loops at the endpoints changes the turn density and leaves large gaps as h grows. The denser model avoids confusing that spacing error with the long-solenoid limit.

The first passing height is h/a = 14.25 on a grid spaced by 0.25. The continuous-winding formula crosses at about 14.04, inside the same bracket. At h/a = 20 the center is close to NI/h, but the field falls near either end. The mouth is about half the center field, and the field keeps dropping outside the coil.

```
Solenoid h/a 14.00 -> center/(NI/h) 0.98994974
Solenoid h/a 14.25 -> center/(NI/h) 0.99029421 (first >= 0.99)
Continuous-winding crossing h/a 14.035848
100 vs 200 turns, normalized threshold change 7.033e-07
Profile h/a = 20: center/(NI/h) 0.995037; mouth/center 0.501867
```





Reversal, scaling, source types, and timing

The source stores Idl as an Mx3 vector that already includes the weights. H must not multiply by those weights again. B uses the existing mu0 function and multiplies H by `mu0*mur`. The tests also check a rectangular current sheet against its solid-angle result, B scaling, reversed curve direction, invalid inputs, and errors at source positions.

```
Current reversal residual 0.000e+00 A/m
Double-current residual 0.000e+00 A/m
Merged elements 800; expected 800
Charge/current merge rejected: em:src:merge:TypeMismatch
5000 observations, 400 elements: 0.197 s (limit 2 s)
```

Interpretation and problems encountered

Lab 3 separated source construction from field evaluation, so the field engine did not need to know whether the elements came from a loop or a sheet. Lab 4 put geometry and weights in one place, but it needed to keep direction as well as length. Keeping those pieces separate made the four new functions short and left charge-source behavior unchanged. For the next interface, it makes sense to keep useful geometry instead of throwing it away just because the first calculation does not need it.

The starter and handout used h differently for the straight wire, so this report states both half-length and full length. The starter also did not sweep the solenoid aspect ratio, and its sparse end-to-end turn spacing did not match N/h closely enough. Two older Lab 6 assertions used MATLAB's max(A,[],'all'), which Octave does not accept. Flattening those arrays keeps the same checks and tolerances. Lab 6 also used the last warning to test potential references; Octave's package warnings could overwrite it, so that check now targets the reference warning directly. The provided runTests and assertClose functions are unchanged.

Full Lab 1 through Lab 9 test output

These results are produced here, not copied from a previous run.

```
EECE 306 toolbox test suite
------------------------------------------------------------
eta = 376.73
  test_lab01 checks complete
  PASS  test_lab01            0.01 s
  test_lab02 checks complete
  PASS  test_lab02            0.02 s
  5000-point field evaluation: 0.0006 s
  test_lab03 checks complete
  PASS  test_lab03            0.02 s
  test_lab04 checks complete
  PASS  test_lab04            0.60 s
  merged source: flux 5.00000000e-09 C, enclosed 5.00000000e-09 C
  density errors at 128000 / 1024000 elements: 0.953% / 1.715% max
  sphere / cube / closed hemisphere: 3.00000000e-09 / 3.00000000e-09 / 3.00000000e-09 C
  sphere Gauss rel err 1.59e-11; midpoint N=960 rel err 4.46e-07
  test_lab05 checks complete
  PASS  test_lab05           20.08 s
  Lab 6: max absolute map error 7.776e-04 V/m
  max closed residual / (max|E| L) 1.252e-13
  Square vs exact-tangent edges 4.177e-11 V
  open midpoint errors [0.0004329 0.0001082 2.706e-05 6.764e-06] V (order 2)
  Fixed line refinement 4.965e-06 V
  growing line potentials 25.9495 -> 37.6527 V
  warning verified
  Step minimum h=3.162e-06 m, error 1.025e-11
  default error 3.586e-11
  10000 points V/grad 0.001/0.007 s
  test_lab06 checks complete
  PASS  test_lab06            1.21 s
  Lab 7: MMS error 8.224e-05, order 2.042
  Jacobi / GS / SOR iterations 1320 / 689 / 77
  Gauss flux 2.5947e-09 C, enclosed 2.5000e-09 C
  test_lab07 checks complete
  PASS  test_lab07            0.36 s
  Lab 8: plate error 8.882e-16; constant-layer error 8.549e-15
  Graded-layer orders harmonic 2.000; arithmetic 0.996
  Coax finest error 1.471%; order 0.983
  Voltage ratio 1.0000000000; filled ratio 3.0000000000
  Bar R 0.01000000 ohm m; tangential E jump 1.776e-14 V/m
  test_lab08 checks complete
  PASS  test_lab08            3.15 s
  Lab 9: max loop relative error 1.040e-11; transverse 1.974e-14 A/m (tol 5.0e-13)
  Wire half-length h/rho 7.25; full-length L/rho 14.50; ratio 0.99062114
  Solenoid full-height h/a 14.25; ratio 0.99029421; mouth/center 0.501867
  H timings: 5000 points 0.099 s; 10000 points 0.197 s
  Surface current, B scaling, reversal, merge, input and singularity checks passed
  test_lab09 checks complete
  PASS  test_lab09            3.01 s
------------------------------------------------------------
  9 passed, 0 failed, 9 total
  All tests passing.
```